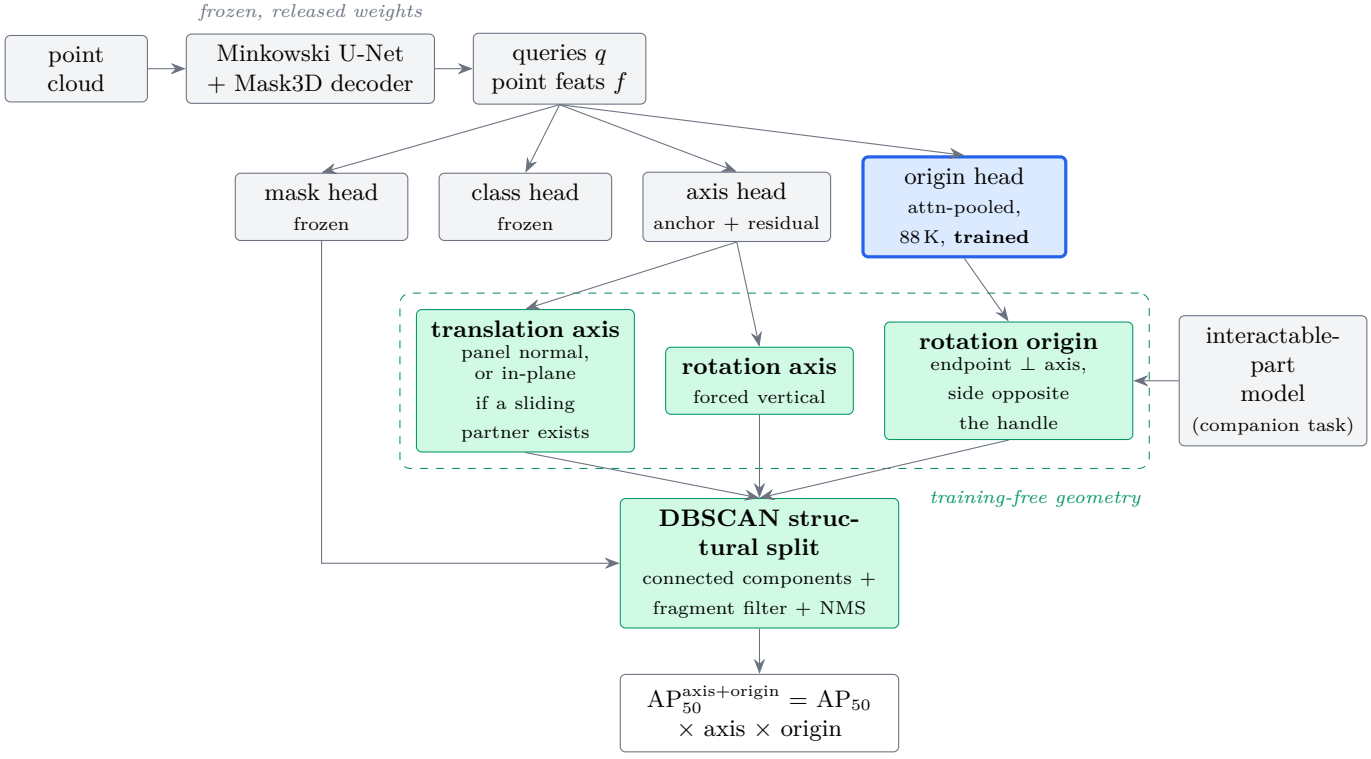
frozen, released weights
point cloud
Minkowski U-Net + Mask3D decoder
queries q point feats f
mask head
frozen
class head
frozen
axis head
anchor + residual
origin head
attn-pooled, 88 K, trained
translation axis
panel normal, or in-plane if a sliding partner exists
rotation axis
forced vertical
rotation origin
endpoint ⊥ axis, side opposite the handle
interactable-part model
(companion task)
training-free geometry
DBSCAN structural split
connected components + fragment filter + NMS
$AP_{50}^{axis+origin} = AP_{50} \times axis \times origin$